7.12 Quiz: Logarithmic and Exponential Functions — Markscheme

Problems 1–2: no calculator. Problems 3–6: calculator allowed.

1. [Maximum mark: 5]

(a) $\ln 8 + \ln 9 = \ln(8 \cdot 9) = \ln 72.$ **A1**

(b) $2 \ln 6 = \ln 6^2 = \ln 36.$ **A1**

$\ln 36 - \ln 4 = \ln \dfrac{36}{4} = \ln 9.$ **A1**

(c) $2 \ln 2 = \ln 4$, so $\ln 48 - 2 \ln 2 - \ln 3 = \ln 48 - \ln 4 - \ln 3.$ **A1**

$= \ln \dfrac{48}{4 \cdot 3} = \ln 4.$ **A1**

Accept any valid ordering of the product/quotient rules, e.g. $\ln 48 - \ln(4 \cdot 3) = \ln(48/12) = \ln 4.$

2. [Maximum mark: 5]

(a) $3^{2x} = 27 = 3^3$, so $2x = 3$. **M1**

$x = \dfrac{3}{2}$. **A1**

(b) Use $\log_2 a + \log_2 b = \log_2(ab)$:

$\log_2\big[(x+3)(x-3)\big] = 4$. **M1**

Exponentiate: $(x+3)(x-3) = 2^4 = 16$, so $x^2 - 9 = 16$. **A1**

$x^2 = 25$. Since $x > 3$: $x = 5$. **A1**

Reject $x = -5$ (outside the stated domain $x > 3$).

3. [Maximum mark: 6]

$g(x) = 2 + \log_3(x - 1)$.

(a) Vertical asymptote where the argument of the log is zero: $x - 1 = 0$, so $x = 1$. **A1**

(b) Set $g(x) = 0$: $\log_3(x - 1) = -2$. **M1**

$x - 1 = 3^{-2} = \dfrac{1}{9}$, so $x = \dfrac{10}{9}$. **A1**

Accept $x \approx 1.11$ *(3 s.f.) or the point* $(\frac{10}{9}, 0)$.

(c) Let $y = 2 + \log_3(x - 1)$. Interchange x and y: $x = 2 + \log_3(y - 1)$. **M1**

$\log_3(y - 1) = x - 2$, so $y - 1 = 3^{x-2}$. **A1**

$g^{-1}(x) = 1 + 3^{x-2}$. **A1**

Accept equivalent forms, e.g. $g^{-1}(x) = 1 + \dfrac{3^x}{9}$.

4. [Maximum mark: 6]

$A(t) = 8\,e^{kt}$, $k > 0$, with $A(4) = 32$.

(a) $8\,e^{4k} = 32$, so $e^{4k} = 4$. **M1**

$4k = \ln 4 = 2\ln 2$. **A1**

$k = \frac{1}{2}\ln 2$. **AG**

(b) $A(11) = 8\,e^{11k} = 8 \cdot 2^{11/2} = 256\sqrt{2}$. **M1**

GDC: $256 \times 1.41421\ldots = 362.038\ldots$

$A(11) \approx 362$ kg (nearest integer). **A1**

(c) Set $8\,e^{kt} = 1000$, so $e^{kt} = 125$.

$kt = \ln 125$, hence $t = \dfrac{\ln 125}{\frac{1}{2}\ln 2} = \dfrac{2\ln 125}{\ln 2}$. **M1**

$t = 13.9316\ldots \approx 13.93$ days (2 d.p.). **A1**

Accept $t = 6\log_2 5$ *as an equivalent exact form; GDC solve of* $8\,e^{(t/2)\ln 2} = 1000$ *also acceptable.*

5. [Maximum mark: 6]

Principal \$5000, nominal annual rate 4.8%, compounded monthly.

(a) $V(t) = 5000\left(1 + \dfrac{0.048}{12}\right)^{12t} = 5000\,(1.004)^{12t}$. **A1**

(b) $V(6) = 5000\,(1.004)^{72}$. **M1**

GDC: $5000 \times 1.332989\ldots = 6664.95$.

$V(6) = \$6664.95$ (nearest cent). **A1**

Accept \$6664.96 *from full-precision evaluation.*

(c) Need the smallest integer n with $5000\,(1.004)^{12n} > 8000$, i.e. $(1.004)^{12n} > 1.6$. **M1**

Take logs: $12n > \dfrac{\ln 1.6}{\ln 1.004} = 117.737\ldots$, so $n > 9.811\ldots$ **A1**

Smallest full-year $n = 10$ years. **A1**

GDC check: $V(9) = 5000\,(1.004)^{108} = \$7695.03$ *(under);* $V(10) = 5000\,(1.004)^{120} = \$8072.62$ *(over). Two-row table earns full marks; single-row table earns full marks with a feedback note recommending the adjacent row.*

6. [Maximum mark: 6]

$L = 10\log_{10}\left(\dfrac{I}{I_0}\right)$, $I_0 = 10^{-12}$ W/m^2.

(a) $\dfrac{I}{I_0} = \dfrac{3.2\times10^{-5}}{10^{-12}} = 3.2\times10^{7}$. **M1**

$L = 10\log_{10}(3.2\times10^{7}) = 10\times7.50515\ldots \approx 75.1$ dB (1 d.p.). **A1**

(b) $130 = 10\log_{10}\left(\dfrac{I}{10^{-12}}\right)$, so $\log_{10}\left(\dfrac{I}{10^{-12}}\right) = 13$. **M1**

$\dfrac{I}{10^{-12}} = 10^{13}$, hence $I = 10$, i.e. $I = 1\times10^{1}$ W/m^2. **A1**

Accept $I = 10$ *W/m^2;* $a = 1$, $k = 1$.

(c) $L_{2I} - L_I = 10\log_{10}\left(\dfrac{2I}{I_0}\right) - 10\log_{10}\left(\dfrac{I}{I_0}\right) = 10\log_{10}2$. **M1**

$10\log_{10}2 = 10\times0.30103\ldots = 3.0103\ldots$ **A1**

≈ 3.01 dB. **AG**